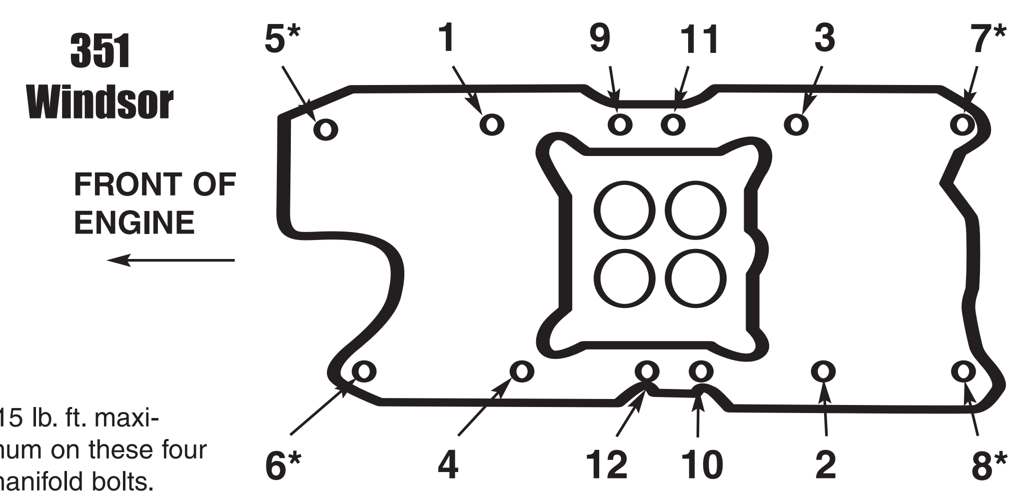
226042 - Small Block Chrysler including Magnum

Application - This manifold will fit 1967 - '91 340 and 360 Chrysler engines, 1967 and later 318 and all 318's using 340 or 360 cylinder heads. It will also fit the 1992 and later Magnum engines.
Warning - This manifold accepts the early style thermostat housing (waterneck) only. It will not accept stock Thermo-Quad carburetor. There is no provision for exhaust heated chokes and no exhaust crossover. 1979 and later rotary A/C compressor will not clear waterneck.
Dowel Pin Removal - Remove the two dowel pins at each end of the block where the manifold sits. Discard them.
Dual Bolt Pattern - You will note that this manifold features two sets of bolt holes; one set at an angle for all installations except 1992 and later Magnum engines, and the vertical bolt holes are for the 1992 and later Magnum engines, or early engines with Magnum heads. We include a set of polished buttons that can be tapped into the set of holes that are not used. This will keep dirt and grease from collecting in these holes and will also provide a more finished appearance to your installation.
Note: Early installations use 3/8" bolts while later Magnum style uses 5/16" bolts. We make all our bolt holes the larger size to accomodate the tap-in buttons. If you are installing this manifold on a Magnun engine, carefully align the manifold over the bolt holes. Look down through the holes in the manifold and make sure that the tapped holes in the head are approximately centered in relationship to the holes in the manifold. Use your original manifold bolts along with the supplied washers in this kit. Torque bolts in sequence shown.
Front Bypass Water Port - There is a pipe tapped hole in the front of the manifold, just under the thermostat housing flange. This port is tapped 3/4-14 NPT. 1970 and later used this size port, earlier engines used 1/2 NPT. If you want to use the early fitting, you will need a 3/4 to 1/2 pipe bushing, available at most hardware stores.
Intake Gaskets - Do not use competition style intake gaskets for street applications. Long term street use will deteriorate these type of gaskets and lead to fuel and oil leakage. Use Fel-Pro Printoseal #1213 for 1967-'91 applications and Mopar #P-4876049 for 1992 and later engines.
Cruise Control Bracket - If your vehicle is equipped with cruise control, use the supplied bracket to move the cruise control clamp which will ensure proper clamping of the cruise control cable.
Hood Clearance - Due to the extra height of this manifold, always check that you have adequate hood clearance before closing the hood the first time.
Carburetor Recommendations:
Holley 0-80457S (600 cfm)
Holley 0-4778S (700 cfm)
Holley 0-80508S (750 cfm)
Edelbrock #1407 (750 cfm)*
Edelbrock #1412 (800 cfm)*
Road Demon #4282010VE (625 cfm)
Road Demon #4402020VE (695 cfm)
Speed Demon #1282010 (650 cfm)
Speed Demon #1402010 (750 cfm)
*Use Edelbrock #1481 Throttle/Trans Lever Kit



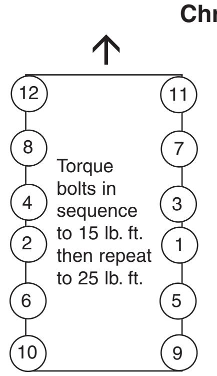
*15 lb. ft. maximum on these four manifold bolts.

MANIFOLD BOLT TIGHTENING SEQUENCE

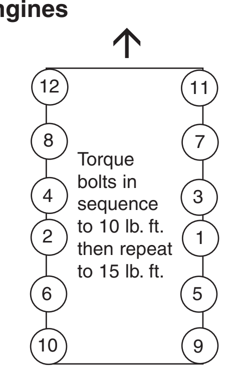
Note: Please follow this tightening sequence and be sure to read instructions given above as damage can occur to manifold if instructions are not carefully followed.

Front of Engine

Chrysler Engines

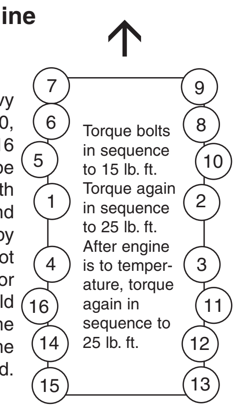


Torque sequence for non-Magnum engines

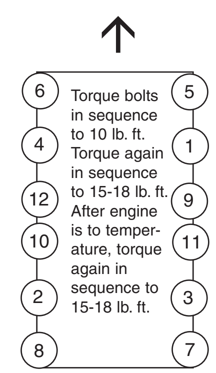


Torque sequence for Magnum engines

On BB Chevy bolts #5, 10, 11 and 16 should be tightened with a box end wrench by hand. Do not overtighten or you could break the ears off the manifold.



Torque sequence for Big Block Chevy



Torque sequence for Small Block Ford

SUMMIT® RACING EQUIPMENT

INTAKE MANIFOLD GENERAL INSTRUCTIONS FOR REMOVAL AND INSTALLATION.

Tech Hotline: 1-330-630-0240

NOTE: PLEASE READ ALL INSTRUCTIONS BEFORE INSTALLATION. See instructions for specific manifolds starting on page 3 of these instructions.

This Instruction sheet is designed to cover a wide variety of vehicle applications. If your vehicle is not equipped with any items referred to in these instructions (EGR, transmission kick-down linkage, air conditioning, power brakes, etc.), proceed to the next step.

Follow these instructions carefully, so that you can achieve the results intended for this high quality performance intake manifold. Slight errors in installation can make a big difference in performance, mileage and emissions. Warranty is void if proper installation procedures are not followed.

CHECK LIST:

- Fully read and understand all of the instructions.
- Inspect manifold for any shipping damage. If damaged, contact Summit immediately.
- All threaded holes have been checked by the factory but do a quick check to make sure they are all clean and not damaged.
- Check all internal passages with a flashlight and a wire, making sure the passages are clean and not obstructed.
- Use proper OEM or aftermarket intake gaskets. See bottom of page 2 of this instruction sheet for specific gasket recommendations.
- Use Teflon® tape on all pipe plugs.
- Remove dowel pins from end seal surface on Ford & Chrysler products. Use vise grip pliers for removal.
- Use correct carburetor and adapter if recommended. Always use a new carburetor gasket.
- Re-install vacuum lines correctly and replace any bad lines with the correct size.
- Position manifold and follow torque sequence correctly per instruction sheet.
- Set ignition timing to correct specifications.
- Test drive your vehicle before installation, noting at what point automatic shift points occur. After installation of this manifold, adjust linkage to achieve same shift points. (If applicable)
- Check emission parts for proper function before removing stock manifold.
- Adjust automatic choke correctly.

NOTE: This manifold was throroughly checked at the factory. It has been pressure tested to assure that no air or water leaks occur. All the tapped holes have been cleaned after polishing with the proper taps. The manifold has been vapor blasted to clean it and then meticulously hand detailed before packing for shipment.

PARTS REQUIRED:

1. Intake manifold gasket set. See bottom of page 2 of these instructions for specific recommendations.
2. Oil resistant, silicone based sealant such as Mr.Gasket Part No. 7809, or equivalent.
3. Spray gasket adhesive, such as Fel Pro's "Spray Tack" Part No. 220, or equivalent.
4. Pipe plugs, If needed, either 3/8 or 1/2-NPT.
5. Carburetor base gasket (usually supplied with carburetor).
6. Teflon tape or pipe dope.

NOTE: Never install tapered (pipe) fittings in an aluminum manifold without Teflon tape or thread damage and/or leakage will likely occur.

TOOLS NORMALLY REQUIRED:

1. Socket wrench set
2. Open end wrenches
3. Box end/flare wrenches (optional)
4. Distributor wrench
5. Ignition wrench set
6. Screwdrivers (Standard and Phillips)
7. Gasket scraper or putty knife
8. Channel lock and hose clamp pliers
9. Torque wrench
10. Timing light and vacuum gauge
11. Drain bucket
12. Rags
13. 3/8-16 or 5/16-18 tap (for cleaning bolt holes)

MANIFOLD REMOVAL PROCEDURE:

WARNING: Do not attempt to remove manifold from a hot engine. Allow the engine time to cool down sufficiently before removal.

1. Disconnect battery ground cable.
2. Tag vacuum and crankcase ventilation hoses leading to air cleaner, if so equipped, making note of routing and connection points. Now remove vacuum and crankcase hoses allowing removal of the air cleaner assembly.
3. Note the routing of remaining vacuum lines from carburetor and intake manifold. After being tagged, remove vacuum lines.
4. Drain radiator by opening drain plug at lower corner of radiator. If no drain plug is present, it may be necessary to remove the lower radiator hose. CAUTION: Coolant may still be hot. Allow engine to cool down before proceeding.
5. Disconnect throttle linkage and springs, transmission kick-down/cruise control (if applicable), and carburetor choke rod.
6. Remove gas cap to relieve pressure from fuel system. Disconnect fuel line at carburetor using flare wrenches. Plug fuel line to prevent fuel leaks. Remove carburetor
7. Tag and disconnect ignition coil and sensor wires. Remove coil and coil bracket, if mounted on manifold.
8. Remove radiator hose, thermostat housing and thermostat.
9. Remove remaining water hoses and fittings from manifold.
10. Remove all manifold vacuum fittings.
11. Remove any remaining brackets from the manifold.
12. Loosen or remove valve covers, if needed, to aid in manifold removal.

IGNITION REMOVAL PROCEDURES:

FOLLOW INSTRUCTIONS CAREFULLY, AS SERIOUS DAMAGE CAN OCCUR WHEN IGNITION IS NOT RE-INSTALLED CORRECTLY.

1. Remove distributor cap.
2. Note position of rotor, then mark the distributor housing in line with the rotor tip.
3. Note position of distributor vacuum canister and place a reference mark on a convenient surface.
4. Note how far points are open or note position of magnetic trigger wheel (if so equipped); if closed, note the distance from point block to cam lobe.
5. Remove distributor. DO NOT rotate engine after removing distributor.
6. Remove the intake manifold-to-cylinder head bolts.
7. Remove intake manifold.

PREPARATION FOR INSTALLING MANIFOLD:

1. Clean all mating surfaces thoroughly.
2. **CAUTION:** To prevent gasket pieces from falling into ports and valley when cleaning old gaskets from head surfaces, insert rags into ports and lay rags in lifter valley. When clean, remove rags carefully. Make sure that all particles that fell on rags are completely removed. Wipe surfaces with lacquer thinner or alcohol soaked rags to remove any oil or grease. **NOTE:** This is a must for proper gasket sealing.

INSTALLING INTAKE MANIFOLD:

1. Apply a thin coat of spray adhesive to the cylinder head side of the intake gasket surface. Lay manifold gasket in place, aligning ports and bolt holes.
2. Apply a 1/4" wide bead of oil-resistant RTV-silicone sealant to the front and rear block sealing surfaces, making sure to overlap manifold gaskets at all four corners. Do not use cork or rubber seals.
3. Carefully position your intake manifold on engine. Make sure that all bolt holes are centered. If manifold must be moved, re-check gasket placement. Install intake bolts. NOTE: Thread sealant should be used on all bolt threads.
4. Torque bolts to torque specifications and in sequence shown on page 4 of these instructions.

INSTALLING FITTING, PLUGS AND STUDS:

1. Install fittings, plugs, carburetor studs and accessory bolts from your original manifold. Use Teflon® tape or pipe dope on all pipe threads.

CAUTION: Do not over-tighten or cross-thread fittings, plugs, studs or bolts in manifold. Damage to threads or cracked mounting boss may result unless caution is used when installing,

2. Install all water sensors and vacuum fittings into manifold.
3. Plug all un-used water and vacuum ports in the manifold with pipe plugs.

NOTE: Use Teflon® tape or pipe dope on all pipe threads. Anti-seize compound is recommended when installing aluminum pipe plugs in an aluminum manifold.

INSTALLING CARBURETOR:

1. Place new carburetor gasket on clean carburetor mounting surface. Do not use any type of sealant on carburetor gasket.
2. Install carburetor.
3. Connect all linkage and throttle springs.
4. Connect all vacuum and fuel lines. Refer to your tags or drawing for correct placement from Manifold Removal Procedures section (step 3).
5. Automatic transmissions only: Adjust kick-down or throttle pressure linkage for proper shift points. Check all linkages, making sure that they all function properly.
6. Re-tighten gas cap.

WIRES, BRACKETS AND VALVE COVERS:

1. If valve covers were removed, re-install with new gaskets.
2. Install coil brackets, coil, wires, and all remaining brackets that were removed from manifold.

INSTALLING THERMOSTAT:

1. Install thermostat and apply silicone sealant on both sides of gasket and place on manifold. Clean thermostat housing of any old gasket material before positioning on gasket. Start bolts by hand, then tighten.
2. Install heater and radiator hoses. NOTE: Check that radiator drain plug is closed before replacing coolant.

INSTALLING IGNITION:

1. Install distributor at this time making sure distributor fully engages the oil pump drive shaft.
2. Check location of rotor and distributor body, making sure your reference marks line up. Refer to Ignition Removal section (steps 2, 3, and 4). Install hold-down clamp and tighten distributor just enough that it still can be rotated by hand. Re-install distributor cap and wires.
3. Connect battery cable.
4. Hook up timing light and start engine; set timing to factory specs, tighten distributor.
5. Check for possible fuel, oil, or coolant leaks and proper choke operation.
6. Install air cleaner.

CAUTION: Check that there is adequate clearance for throttle and choke linkages through their range of travel.
IMPORTANT: Check for adequate hood clearance before closing hood. Clearance should be 0.5" min.

7. Operate engine for 30 minutes. Allow engine to cool and re-torque manifold bolts.

GENERAL INFORMATION:

1. Periodically (every six months or 6000 miles) re-check the torque on the manifold bolts to minimize the possibility of a vacuum leak.
2. If the cylinder heads have been milled or the cylinder block "decked," the cylinder head faces and the end surfaces of the manifold must be milled to compensate. This is necessary to maintain correct port alignment, minimize the possibility of manifold vacuum leaks, and assure proper engine performance.
3. Ignition timing should be set to factory specifications. Any attempt to further advance the initial ignition setting will result in an adverse effect on exhaust emission levels and improper engine operation. Since idle speed increases as the ignition is advanced, the only way to bring the idle speed down to an acceptable level, is to close the throttle plates with the idle speed adjustment screw. Closing the throttle plates in this manner will change the geometry between the throttle plates and the idle fuel ports. This can cause idle quality deterioration and make it difficult to get the idle mixture rich enough. If more advance is desired, it should be done in the distributor advance curve.
4. If changing from a 2BBL intake manifold (NOTE:

Intake Gasket Recommendations:

226022 & 226020 - Fel-Pro #1210 or #1212
226031 - Fel-Pro #1250
226036 - Fel-Pro #1250
226042 - Fel-Pro #1213 (non-Magnum) or Mopar #P-4876049 (Magnum)

Check legality in your state), it is sometimes necessary to adjust the transmission kick-down linkage to the carburetor in order to obtain wide-open throttle. This adjustment is made by loosening the locking grommet and pulling enough kick-down cable through the grommet to achieve full throttle. Lock the grommet against the kick-down cable and connect kick-down linkage to the carburetor.

INSTRUCTIONS FOR SPECIFIC MANIFOLDS

226022 - Big Block Chevy V8 (Oval Port)

1. Street legal - This manifold is a stock replacement part when used with a stock or legal replacement carburetor on 1967-72 396, 402, 427, 454 V8's and 1966-83 trucks originally sold without EGR. This manifold will not accept stock EGR equipment.
2. This manifold will accept all stock parts (in most cases) when used with the stock carb or one of the carburetors recommended below.
3. Manifold Port Design - This manifold was designed to generate a maximum power band from idle to 6,000 rpm with the latest style oval port heads. We recommend using the manifold as-cast with the early style (larger oval port) cylinder heads when operating within the recommended rpm range.
4. Intake Gaskets - Use Fel-Pro #1210 or #1212.
5. Carburetor recommendations:
   a. Original Quadrajet 4BBL spread bore
   b. Holley #0-80770 770 cfm spread bore
   c. Holley #0-80508S 750 cfm
   d. Holley #0-4779C 750 cfm
   e. Edlebrock #1405, #1406 600 cfm
   f. Edelbrock #1407 750 cfm
   g. Edelbrock Q-Jet #1901, #1902 750 cfm
   h. Edelbrock Q-Jet #1903, #1904 795 cfm
6. This manifold is supplied with a special carb adapter plate and gasket. You will need to install this adapter if you elect to use a square bore carburetor instead of the stock style spread bore.

226020 - Big Block Chevy air gap style - Oval Port

1. An ideal street manifold. Designed for 396-454 engines with oval ports, this manifold will accept late model waterneck, A/C, alternator and HEI ignitions.
2. Carburetor recommendations:
   a. Edelbrock #1407, #1411 750 cfm
   b. Edelbrock #1412, #1413 800 cfm
   c. Holley #0-80508S 750 cfm
   d. Holley #0-4780C 800 cfm
3. Use a Fel-Pro #1212 intake gasket.
4. For high performance, the carburetor pad on this manifold is forward of the stock position by .25". If your throttle bracket fastens to the carb, it is ok as is. If it fastens to back of manifold, use an Edelbrock #8012 Extension Kit. To install, press throttle pedal fully, install the #8012 extension to the carb. Failure to do this will negatively affect trans kickdown and up-shift points.
5. There are four manifold bolts that stick out towards the valve cover further than others. There is no gasket support under these bolt holes and over-tightening them can break the manifold. Hand tighten with a 6" box end wrench. Do not use a torque wrench.

226031 - Air gap style for 260-289-302 SB Ford V8

1. Street Legal - These manifolds are a stock replacment part when used with the OEM carburetor in 1965-66 289 Shelby Mustangs. This manifold will not accept stock EGR equipment.
2. This manifold will accept all stock parts (in most cases) when used with the stock carb or one of the carbs recommended below.
3. Intake Gaskets - Use Fel-Pro #1250 or OE equivalent.
4. Carburetor recommendations:
   a. OEM 4BBL
   b. Holley #0-4779 750 cfm
   c. Edelbrock #1405 600 cfm
   d. Edelbrock #1407 750 cfm
5. Clearances - In some applications with an Edelbrock or Holley carb, there may be some interference with the stock air cleaner. Use an Edelbrock Air Cleaner #1221 or adapter #8092. **Caution:** Check hood clearance when using a spacer.
6. Brackets - A throttle mounting adapter plate is supplied with each manifold to adapt both early and later type stock throttle brackets to our manifold. Our bracket bolts to the manifold with the slotted hole towards the center of the manifold and then the stock throttle bracket bolts to our bracket.
7. Torque specs. See sequence chart on page 4. Following this sequence, torque bolts to 10 lb. ft. Then following same sequence, torque bolts to 15-18 lb. ft. Do not over tighten as damage to the manifold may occur.

226036 - 351W - Air gap style manifold

Special Notes on Early Windsors

Summit 351W intake manifolds have a 12-bolt intake manifold pattern, but they can be used on 1974 and earlier small-block Ford Windsor heads having a 16-bolt pattern. When using our manifold on 16-bolt heads, you have to use a 16-bolt intake gasket to ensure proper sealing around the water ports on the heads.

Recommended Intake Gasket

Due to various changes in Ford cylinder heads, not all intake gaskets will work with this manifold. To assure that you have the right gasket, we recommend the use of the Fel-Pro #1250. Use of other gaskets may result in water leaks. This gasket with this manifold will fit any 351W head.

EGR Systems

This manifold will not accept stock EGR equipment. EGR systems are used on some 1972 and later vehicles but only in some states. Check local laws for requirements.

Manifold Bolt Tightening Procedure

Some heads have provision for 12 manifold bolts and some use 16 bolts. This manifold uses 12 bolts. Install all 12 bolts and torque to 10 lb. ft. Follow torque sequence shown below. Then using same sequence, tighten to 15 lb. ft. Again, following torque sequence, tighten ONLY THE INSIDE 8 BOLTS to 25 lb. ft. Note that the corner four bolts (numbers 5, 6, 7, and 8) are not supported by a cylinder head and tightening them beyond 15 lb. ft. could result in cracking or breaking the manifold.

Carburetor/Throttle Body Recommendations

This manifold is designed to be used with any standard square bore carburetor including Holley, Demon, Edelbrock, or Carter.

Street Performance:

Holley #0-80457S 600 cfm
Holley #0-4777S 700 cfm
Edelbrock #1407 750 cfm

Special Features

These manifolds have several special features not found on the stock manifold or most aftermarket manifolds. One feature is the addition of two tapped water ports in the back portion of the manifold. These holes are 3/8-NPT and must be plugged with pipe plugs if not used. Or you may find it more advantageous to position your temperature gauge sending unit in one of these back ports to avoid running wires to the front of the engine.